

Example 2.32. In the case of S^1 , the map $f(z) = z^k$, where we view S^1 as the unit circle in $\mathbb{C}$, has degree k . This is evident in the case $k = 0$ since f is then constant. The case $k < 0$ reduces to the case $k > 0$ by composing with $z \mapsto z^{-1}$, which is a reflection, of degree -1 . To compute the degree when $k > 0$, observe first that for any $y \in S^1$, $f^{-1}(y)$ consists of k points $x_1, \dots, x_k$ near each of which f is a local homeomorphism, stretching a circular arc by a factor of k . This local stretching can be eliminated by a deformation of f near x_i that does not change local degree, so the local degree at x_i is the same as for a rotation of S^1 . A rotation is a homeomorphism so its local degree at any point equals its global degree, which is $+1$ since a rotation is homotopic to the identity. Hence $\deg f|_{x_i} = 1$ and $\deg f = k$.

Another way of obtaining a map $S^n \rightarrow S^n$ of degree k is to take a repeated suspension of the map $z \mapsto z^k$ in Example 2.32, since suspension preserves degree:

Proposition 2.33. $\deg Sf = \deg f$, where $Sf: S^{n+1} \rightarrow S^{n+1}$ is the suspension of the map $f: S^n \rightarrow S^n$.

Proof: Let CS^n denote the cone $(S^n \times I)/(S^n \times 1)$ with base $S^n = S^n \times 0 \subset CS^n$, so CS^n/S^n is the suspension of S^n . The map f induces $Cf: (CS^n, S^n) \rightarrow (CS^n, S^n)$ with quotient Sf . The naturality of the boundary maps in the long exact sequence of the pair (CS^n, S^n) then gives commutativity of the diagram at the right. Hence if f_* is multiplication by d , so is Sf_* . $\square$

$$\begin{array}{ccc} \tilde{H}_{n+1}(S^{n+1}) & \xrightarrow{\partial} & \tilde{H}_n(S^n) \\ \downarrow Sf_* & & \downarrow f_* \\ \tilde{H}_{n+1}(S^{n+1}) & \xrightarrow{\partial} & \tilde{H}_n(S^n) \end{array}$$

Note that for $f: S^n \rightarrow S^n$, the suspension Sf maps only one point to each of the two ‘poles’ of S^{n+1} . This implies that the local degree of Sf at each pole must equal the global degree of Sf . Thus the local degree of a map $S^n \rightarrow S^n$ can be any integer if $n \geq 2$, just as the degree itself can be any integer when $n \geq 1$.

Cellular Homology

Cellular homology is a very efficient tool for computing the homology groups of CW complexes, based on degree calculations. Before giving the definition of cellular homology, we first establish a few preliminary facts:

Lemma 2.34. *If X is a CW complex, then:*

- (a) $H_k(X^n, X^{n-1})$ is zero for $k \neq n$ and is free abelian for $k = n$, with a basis in one-to-one correspondence with the n -cells of X .
- (b) $H_k(X^n) = 0$ for $k > n$. In particular, if X is finite-dimensional then $H_k(X) = 0$ for $k > \dim X$.
- (c) The inclusion $i: X^n \hookrightarrow X$ induces an isomorphism $i_*: H_k(X^n) \rightarrow H_k(X)$ if $k < n$.

Proof: Statement (a) follows immediately from the observation that (X^n, X^{n-1}) is a good pair and X^n/X^{n-1} is a wedge sum of n -spheres, one for each n -cell of X . Here we are using Proposition 2.22 and Corollary 2.25.

To prove (b), consider the long exact sequence of the pair (X^n, X^{n-1}) , which contains the segments

$$H_{k+1}(X^n, X^{n-1}) \rightarrow H_k(X^{n-1}) \rightarrow H_k(X^n) \rightarrow H_k(X^n, X^{n-1})$$

If k is not equal to n or $n-1$ then the outer two groups are zero by part (a), so we have isomorphisms $H_k(X^{n-1}) \approx H_k(X^n)$ for $k \neq n, n-1$. Thus if $k > n$ we have $H_k(X^n) \approx H_k(X^{n-1}) \approx H_k(X^{n-2}) \approx \dots \approx H_k(X^0) = 0$, proving (b). Further, if $k < n$ then $H_k(X^n) \approx H_k(X^{n+1}) \approx \dots \approx H_k(X^{n+m})$ for all $m \geq 0$, proving (c) if X is finite-dimensional.

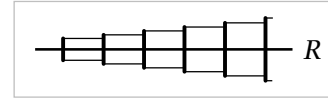
The proof of (c) when X is infinite-dimensional requires more work, and this can be done in two different ways. The more direct approach is to descend to the chain level and use the fact that a singular chain in X has compact image, hence meets only finitely many cells of X by Proposition A.1 in the Appendix. Thus each chain lies in a finite skeleton X^m . So a k -cycle in X is a cycle in some X^m , and then by the finite-dimensional case of (c), the cycle is homologous to a cycle in X^n if $n > k$, so $i_*: H_k(X^n) \rightarrow H_k(X)$ is surjective. Similarly for injectivity, if a k -cycle in X^n bounds a chain in X , this chain lies in some X^m with $m \geq n$, so by the finite-dimensional case the cycle bounds a chain in X^n if $n > k$.

The other approach is more general. From the long exact sequence of the pair (X, X^n) it suffices to show $H_k(X, X^n) = 0$ for $k \leq n$. Since $H_k(X, X^n) \approx \tilde{H}_k(X/X^n)$, this reduces the problem to showing:

(*) $\tilde{H}_k(X) = 0$ for $k \leq n$ if the n -skeleton of X is a point.

When X is finite-dimensional, (*) is immediate from the finite-dimensional case of (c) which we have already shown. It will suffice therefore to reduce the infinite-dimensional case to the finite-dimensional case. This reduction will be achieved by stretching X out to a complex that is at least locally finite-dimensional, using a special case of the ‘mapping telescope’ construction described in greater generality in §3.F.

Consider $X \times [0, \infty)$ with its product cell structure, where we give $[0, \infty)$ the cell structure with the integer points as 0-cells. Let $T = \bigcup_i X^i \times [i, \infty)$, a subcomplex of $X \times [0, \infty)$.



The figure shows a schematic picture of T with $[0, \infty)$ in the horizontal direction and the subcomplexes $X^i \times [i, i+1]$ as rectangles whose size increases with i since $X^i \subset X^{i+1}$. The line labeled R can be ignored for now. We claim that $T \simeq X$, hence $H_k(X) \approx H_k(T)$ for all k . Since X is a deformation retract of $X \times [0, \infty)$, it suffices to show that $X \times [0, \infty)$ also deformation retracts onto T . Let $Y_i = T \cup (X \times [i, \infty))$. Then Y_i deformation retracts onto Y_{i+1} since $X \times [i, i+1]$ deformation retracts onto $X^i \times [i, i+1] \cup X \times \{i+1\}$ by Proposition 0.16. If we perform the deformation retraction of Y_i onto Y_{i+1} during the t -interval $[1 - 1/2^i, 1 - 1/2^{i+1}]$, then this gives a deformation retraction f_t of $X \times [0, \infty)$ onto T , with points in $X^i \times [0, \infty)$ stationary under f_t for $t \geq 1 - 1/2^{i+1}$. Continuity follows from the fact

that CW complexes have the weak topology with respect to their skeleta, so a map is continuous if its restriction to each skeleton is continuous.

Recalling that X^0 is a point, let $R \subset T$ be the ray $X^0 \times [0, \infty)$ and let $Z \subset T$ be the union of this ray with all the subcomplexes $X^i \times \{i\}$. Then Z/R is homeomorphic to $\bigvee_i X^i$, a wedge sum of finite-dimensional complexes with n -skeleton a point, so the finite-dimensional case of (*) together with Corollary 2.25 describing the homology of wedge sums implies that $\tilde{H}_k(Z/R) = 0$ for $k \leq n$. The same is therefore true for Z , from the long exact sequence of the pair (Z, R) , since R is contractible. Similarly, T/Z is a wedge sum of finite-dimensional complexes with $(n+1)$ -skeleton a point, since if we first collapse each subcomplex $X^i \times \{i\}$ of T to a point, we obtain the infinite sequence of suspensions SX^i ‘skewed’ along the ray R , and then if we collapse R to a point we obtain $\bigvee_i \Sigma X^i$ where ΣX^i is the reduced suspension of X^i , obtained from SX^i by collapsing the line segment $X^0 \times [i, i+1]$ to a point, so ΣX^i has $(n+1)$ -skeleton a point. Thus $\tilde{H}_k(T/Z) = 0$ for $k \leq n+1$, and then the long exact sequence of the pair (T, Z) implies that $\tilde{H}_k(T) = 0$ for $k \leq n$, and we have proved (*). $\square$

Let X be a CW complex. Using Lemma 2.34, portions of the long exact sequences for the pairs (X^{n+1}, X^n) , (X^n, X^{n-1}) , and (X^{n-1}, X^{n-2}) fit into a diagram

$$\begin{array}{ccccccc}
 & & & & & & 0 \\
 & & & & & \nearrow & \\
 & & & & H_n(X^{n+1}) \approx H_n(X) & & \\
 & & 0 & \searrow & \nearrow & & \\
 & & & H_n(X^n) & & & \\
 \partial_{n+1} \nearrow & & & \searrow j_n & & & \\
 \dots \longrightarrow H_{n+1}(X^{n+1}, X^n) & \xrightarrow{d_{n+1}} & H_n(X^n, X^{n-1}) & \xrightarrow{d_n} & H_{n-1}(X^{n-1}, X^{n-2}) & \longrightarrow \dots \\
 & & \searrow \partial_n & \nearrow j_{n-1} & & & \\
 & & & H_{n-1}(X^{n-1}) & & & \\
 & & 0 \nearrow & & & &
 \end{array}$$

where d_{n+1} and d_n are defined as the compositions $j_n \partial_{n+1}$ and $j_{n-1} \partial_n$, which are just ‘relativizations’ of the boundary maps ∂_{n+1} and ∂_n . The composition $d_n d_{n+1}$ includes two successive maps in one of the exact sequences, hence is zero. Thus the horizontal row in the diagram is a chain complex, called the **cellular chain complex** of X since $H_n(X^n, X^{n-1})$ is free with basis in one-to-one correspondence with the n -cells of X , so one can think of elements of $H_n(X^n, X^{n-1})$ as linear combinations of n -cells of X . The homology groups of this cellular chain complex are called the **cellular homology groups** of X . Temporarily we denote them $H_n^{CW}(X)$.

Theorem 2.35. $H_n^{CW}(X) \approx H_n(X)$.

Proof: From the diagram above, $H_n(X)$ can be identified with $H_n(X^n)/\text{Im } \partial_{n+1}$. Since j_n is injective, it maps $\text{Im } \partial_{n+1}$ isomorphically onto $\text{Im}(j_n \partial_{n+1}) = \text{Im } d_{n+1}$

and $H_n(X^n)$ isomorphically onto $\text{Im } j_n = \text{Ker } \partial_n$. Since j_{n-1} is injective, $\text{Ker } \partial_n = \text{Ker } d_n$. Thus j_n induces an isomorphism of the quotient $H_n(X^n)/\text{Im } \partial_{n+1}$ onto $\text{Ker } d_n/\text{Im } d_{n+1}$. $\square$

Here are a few immediate applications:

- (i) $H_n(X) = 0$ if X is a CW complex with no n -cells.
- (ii) More generally, if X is a CW complex with k n -cells, then $H_n(X)$ is generated by at most k elements. For since $H_n(X^n, X^{n-1})$ is free abelian on k generators, the subgroup $\text{Ker } d_n$ must be generated by at most k elements, hence also the quotient $\text{Ker } d_n/\text{Im } d_{n+1}$.
- (iii) If X is a CW complex having no two of its cells in adjacent dimensions, then $H_n(X)$ is free abelian with basis in one-to-one correspondence with the n -cells of X . This is because the cellular boundary maps d_n are automatically zero in this case.

This last observation applies for example to $\mathbb{CP}^n$, which has a CW structure with one cell of each even dimension $2k \leq 2n$ as we saw in Example 0.6. Thus

$$H_i(\mathbb{CP}^n) \approx \begin{cases} \mathbb{Z} & \text{for } i = 0, 2, 4, \dots, 2n \\ 0 & \text{otherwise} \end{cases}$$

Another simple example is $S^n \times S^n$ with $n > 1$, using the product CW structure consisting of a 0-cell, two n -cells, and a $2n$ -cell.

It is possible to prove the statements (i)–(iii) for finite-dimensional CW complexes by induction on the dimension, without using cellular homology but only the basic results from the previous section. However, the viewpoint of cellular homology makes (i)–(iii) quite transparent.

Next we describe how the cellular boundary maps d_n can be computed. When $n = 1$ this is easy since the boundary map $d_1 : H_1(X^1, X^0) \rightarrow H_0(X^0)$ is the same as the simplicial boundary map $\Delta_1(X) \rightarrow \Delta_0(X)$. In case X is connected and has only one 0-cell, then d_1 must be 0, otherwise $H_0(X)$ would not be $\mathbb{Z}$. When $n > 1$ we will show that d_n can be computed in terms of degrees:

Cellular Boundary Formula. $d_n(e_\alpha^n) = \sum_\beta d_{\alpha\beta} e_\beta^{n-1}$ where $d_{\alpha\beta}$ is the degree of the map $S_\alpha^{n-1} \rightarrow X^{n-1} \rightarrow S_\beta^{n-1}$ that is the composition of the attaching map of e_α^n with the quotient map collapsing $X^{n-1} - e_\beta^{n-1}$ to a point.

Here we are identifying the cells e_α^n and e_β^{n-1} with generators of the corresponding summands of the cellular chain groups. The summation in the formula contains only finitely many terms since the attaching map of e_α^n has compact image, so this image meets only finitely many cells e_β^{n-1} .

To derive the cellular boundary formula, consider the commutative diagram

$$\begin{array}{ccccc}
H_n(D_\alpha^n, \partial D_\alpha^n) & \xrightarrow[\approx]{\partial} & \tilde{H}_{n-1}(\partial D_\alpha^n) & \xrightarrow{\Delta_{\alpha\beta*}} & \tilde{H}_{n-1}(S_\beta^{n-1}) \\
\downarrow \Phi_{\alpha*} & & \downarrow \varphi_{\alpha*} & & \uparrow q_{\beta*} \\
H_n(X^n, X^{n-1}) & \xrightarrow{\partial_n} & \tilde{H}_{n-1}(X^{n-1}) & \xrightarrow{q_*} & \tilde{H}_{n-1}(X^{n-1}/X^{n-2}) \\
\searrow d_n & & \downarrow j_{n-1} & & \downarrow \approx \\
& & H_{n-1}(X^{n-1}, X^{n-2}) & \xrightarrow[\approx]{} & H_{n-1}(X^{n-1}/X^{n-2}, X^{n-2}/X^{n-2})
\end{array}$$

where:

- Φ_α is the characteristic map of the cell e_α^n and φ_α is its attaching map.
- $q: X^{n-1} \rightarrow X^{n-1}/X^{n-2}$ is the quotient map.
- $q_\beta: X^{n-1}/X^{n-2} \rightarrow S_\beta^{n-1}$ collapses the complement of the cell e_β^{n-1} to a point, the resulting quotient sphere being identified with $S_\beta^{n-1} = D_\beta^{n-1}/\partial D_\beta^{n-1}$ via the characteristic map Φ_β .
- $\Delta_{\alpha\beta}: \partial D_\alpha^n \rightarrow S_\beta^{n-1}$ is the composition $q_\beta q \varphi_\alpha$, in other words, the attaching map of e_α^n followed by the quotient map $X^{n-1} \rightarrow S_\beta^{n-1}$ collapsing the complement of e_β^{n-1} in X^{n-1} to a point.

The map $\Phi_{\alpha*}$ takes a chosen generator $[D_\alpha^n] \in H_n(D_\alpha^n, \partial D_\alpha^n)$ to a generator of the $\mathbb{Z}$ summand of $H_n(X^n, X^{n-1})$ corresponding to e_α^n . Letting e_α^n denote this generator, commutativity of the left half of the diagram then gives $d_n(e_\alpha^n) = j_{n-1} \varphi_{\alpha*} \partial[D_\alpha^n]$. In terms of the basis for $H_{n-1}(X^{n-1}, X^{n-2})$ corresponding to the cells e_β^{n-1} , the map $q_{\beta*}$ is the projection of $H_{n-1}(X^{n-1}/X^{n-2})$ onto its $\mathbb{Z}$ summand corresponding to e_β^{n-1} . Commutativity of the diagram then yields the formula for d_n given above.

Example 2.36. Let M_g be the closed orientable surface of genus g with its usual CW structure consisting of one 0-cell, $2g$ 1-cells, and one 2-cell attached by the product of commutators $[a_1, b_1] \cdots [a_g, b_g]$. The associated cellular chain complex is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d_2} \mathbb{Z}^{2g} \xrightarrow{d_1} \mathbb{Z} \longrightarrow 0$$

As observed above, d_1 must be 0 since there is only one 0-cell. Also, d_2 is 0 because each a_i or b_i appears with its inverse in $[a_1, b_1] \cdots [a_g, b_g]$, so the maps $\Delta_{\alpha\beta}$ are homotopic to constant maps. Since d_1 and d_2 are both zero, the homology groups of M_g are the same as the cellular chain groups, namely, $\mathbb{Z}$ in dimensions 0 and 2, and $\mathbb{Z}^{2g}$ in dimension 1.

Example 2.37. The closed nonorientable surface N_g of genus g has a cell structure with one 0-cell, g 1-cells, and one 2-cell attached by the word $a_1^2 a_2^2 \cdots a_g^2$. Again $d_1 = 0$, and $d_2: \mathbb{Z} \rightarrow \mathbb{Z}^g$ is specified by the equation $d_2(1) = (2, \dots, 2)$ since each a_i appears in the attaching word of the 2-cell with total exponent 2, which means that each $\Delta_{\alpha\beta}$ is homotopic to the map $z \mapsto z^2$, of degree 2. Since $d_2(1) = (2, \dots, 2)$, we have d_2 injective and hence $H_2(N_g) = 0$. If we change the basis for $\mathbb{Z}^g$ by replacing the last standard basis element $(0, \dots, 0, 1)$ by $(1, \dots, 1)$, we see that $H_1(N_g) \approx \mathbb{Z}^{g-1} \oplus \mathbb{Z}_2$.

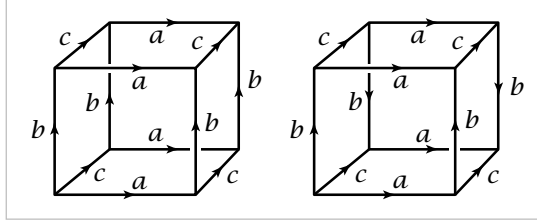
These two examples illustrate the general fact that the orientability of a closed connected manifold M of dimension n is detected by $H_n(M)$, which is $\mathbb{Z}$ if M is orientable and 0 otherwise. This is shown in Theorem 3.26.

Example 2.38: An Acyclic Space. Let X be obtained from $S^1 \vee S^1$ by attaching two 2-cells by the words a^5b^{-3} and $b^3(ab)^{-2}$. Then $d_2: \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$ has matrix $\begin{pmatrix} 5 & -2 \\ -3 & 1 \end{pmatrix}$, with the two columns coming from abelianizing a^5b^{-3} and $b^3(ab)^{-2}$ to $5a - 3b$ and $-2a + b$, in additive notation. The matrix has determinant -1 , so d_2 is an isomorphism and $\tilde{H}_i(X) = 0$ for all i . Such a space X is called **acyclic**.

We can see that this acyclic space is not contractible by considering $\pi_1(X)$, which has the presentation $\langle a, b \mid a^5b^{-3}, b^3(ab)^{-2} \rangle$. There is a nontrivial homomorphism from this group to the group G of rotational symmetries of a regular dodecahedron, sending a to the rotation ρ_a through angle $2\pi/5$ about the axis through the center of a pentagonal face, and b to the rotation ρ_b through angle $2\pi/3$ about the axis through a vertex of this face. The composition $\rho_a\rho_b$ is a rotation through angle π about the axis through the midpoint of an edge abutting this vertex. Thus the relations $a^5 = b^3 = (ab)^2$ defining $\pi_1(X)$ become $\rho_a^5 = \rho_b^3 = (\rho_a\rho_b)^2 = 1$ in G , which means there is a well-defined homomorphism $\rho: \pi_1(X) \rightarrow G$ sending a to ρ_a and b to ρ_b . It is not hard to see that G is generated by ρ_a and ρ_b , so ρ is surjective. With more work one can compute that the kernel of ρ is $\mathbb{Z}_2$, generated by the element $a^5 = b^3 = (ab)^2$, and this $\mathbb{Z}_2$ is in fact the center of $\pi_1(X)$. In particular, $\pi_1(X)$ has order 120 since G has order 60.

After these 2-dimensional examples, let us now move up to three dimensions, where we have the additional task of computing the cellular boundary map d_3 .

Example 2.39. A 3-dimensional torus $T^3 = S^1 \times S^1 \times S^1$ can be constructed from a cube by identifying each pair of opposite square faces as in the first of the two figures. The second figure shows a slightly different pattern of



identifications of opposite faces, with the front and back faces now identified via a rotation of the cube around a horizontal left-right axis. The space produced by these identifications is the product $K \times S^1$ of a Klein bottle and a circle. For both T^3 and $K \times S^1$ we have a CW structure with one 3-cell, three 2-cells, three 1-cells, and one 0-cell. The cellular chain complexes thus have the form

$$0 \rightarrow \mathbb{Z} \xrightarrow{d_3} \mathbb{Z}^3 \xrightarrow{d_2} \mathbb{Z}^3 \xrightarrow{0} \mathbb{Z} \rightarrow 0$$

In the case of the 3-torus T^3 the cellular boundary map d_2 is zero by the same calculation as for the 2-dimensional torus. We claim that d_3 is zero as well. This amounts to saying that the three maps $\Delta_{\alpha\beta}: S^2 \rightarrow S^2$ corresponding to the three 2-cells

have degree zero. Each $\Delta_{\alpha\beta}$ maps the interiors of two opposite faces of the cube homeomorphically onto the complement of a point in the target S^2 and sends the remaining four faces to this point. Computing local degrees at the center points of the two opposite faces, we see that the local degree is $+1$ at one of these points and -1 at the other, since the restrictions of $\Delta_{\alpha\beta}$ to these two faces differ by a reflection of the boundary of the cube across the plane midway between them, and a reflection has degree -1 . Since the cellular boundary maps are all zero, we deduce that $H_i(T^3)$ is $\mathbb{Z}$ for $i = 0, 3$, $\mathbb{Z}^3$ for $i = 1, 2$, and 0 for $i > 3$.

For $K \times S^1$, when we compute local degrees for the front and back faces we find that the degrees now have the same rather than opposite signs since the map $\Delta_{\alpha\beta}$ on these two faces differs not by a reflection but by a rotation of the boundary of the cube. The local degrees for the other faces are the same as before. Using the letters A, B, C to denote the 2-cells given by the faces orthogonal to the edges a, b, c , respectively, we have the boundary formulas $d_3 e^3 = 2C$, $d_2 A = 2b$, $d_2 B = 0$, and $d_2 C = 0$. It follows that $H_3(K \times S^1) = 0$, $H_2(K \times S^1) = \mathbb{Z} \oplus \mathbb{Z}_2$, and $H_1(K \times S^1) = \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}_2$.

Many more examples of a similar nature, quotients of a cube or other polyhedron with faces identified in some pattern, could be worked out in similar fashion. But let us instead turn to some higher-dimensional examples.

Example 2.40: Moore Spaces. Given an abelian group G and an integer $n \geq 1$, we will construct a CW complex X such that $H_n(X) \approx G$ and $\tilde{H}_i(X) = 0$ for $i \neq n$. Such a space is called a **Moore space**, commonly written $M(G, n)$ to indicate the dependence on G and n . It is probably best for the definition of a Moore space to include the condition that $M(G, n)$ be simply-connected if $n > 1$. The spaces we construct will have this property.

As an easy special case, when $G = \mathbb{Z}_m$ we can take X to be S^n with a cell e^{n+1} attached by a map $S^n \rightarrow S^n$ of degree m . More generally, any finitely generated G can be realized by taking wedge sums of examples of this type for finite cyclic summands of G , together with copies of S^n for infinite cyclic summands of G .

In the general nonfinitely generated case let $F \rightarrow G$ be a homomorphism of a free abelian group F onto G , sending a basis for F onto some set of generators of G . The kernel K of this homomorphism is a subgroup of a free abelian group, hence is itself free abelian. Choose bases $\{x_\alpha\}$ for F and $\{y_\beta\}$ for K , and write $y_\beta = \sum_\alpha d_{\beta\alpha} x_\alpha$. Let $X^n = \bigvee_\alpha S_\alpha^n$, so $H_n(X^n) \approx F$ via Corollary 2.25. We will construct X from X^n by attaching cells e_β^{n+1} via maps $f_\beta: S^n \rightarrow X^n$ such that the composition of f_β with the projection onto the summand S_α^n has degree $d_{\beta\alpha}$. Then the cellular boundary map d_{n+1} will be the inclusion $K \hookrightarrow F$, hence X will have the desired homology groups.

The construction of f_β generalizes the construction in Example 2.31 of a map $S^n \rightarrow S^n$ of given degree. Namely, we can let f_β map the complement of $\sum_\alpha |d_{\beta\alpha}|$

disjoint balls in S^n to the 0-cell of X^n while sending $|d_{\beta\alpha}|$ of the balls onto the summand S_α^n by maps of degree $+1$ if $d_{\beta\alpha} > 0$, or degree -1 if $d_{\beta\alpha} < 0$.

Example 2.41. By taking a wedge sum of the Moore spaces constructed in the preceding example for varying n we obtain a connected CW complex with any prescribed sequence of homology groups in dimensions $1, 2, 3, \dots$.

Example 2.42: Real Projective Space $\mathbb{RP}^n$. As we saw in Example 0.4, $\mathbb{RP}^n$ has a CW structure with one cell e^k in each dimension $k \leq n$, and the attaching map for e^k is the 2-sheeted covering projection $\varphi: S^{k-1} \rightarrow \mathbb{RP}^{k-1}$. To compute the boundary map d_k we compute the degree of the composition $S^{k-1} \xrightarrow{\varphi} \mathbb{RP}^{k-1} \xrightarrow{q} \mathbb{RP}^{k-1}/\mathbb{RP}^{k-2} = S^{k-1}$, with q the quotient map. The map $q\varphi$ is a homeomorphism when restricted to each component of $S^{k-1} - S^{k-2}$, and these two homeomorphisms are obtained from each other by precomposing with the antipodal map of S^{k-1} , which has degree $(-1)^k$. Hence $\deg q\varphi = \deg \mathbb{1} + \deg(-\mathbb{1}) = 1 + (-1)^k$, and so d_k is either 0 or multiplication by 2 according to whether k is odd or even. Thus the cellular chain complex for $\mathbb{RP}^n$ is

$$\begin{aligned} 0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \dots \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0 & \quad \text{if } n \text{ is even} \\ 0 \rightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \dots \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0 & \quad \text{if } n \text{ is odd} \end{aligned}$$

From this it follows that

$$H_k(\mathbb{RP}^n) = \begin{cases} \mathbb{Z} & \text{for } k = 0 \text{ and for } k = n \text{ odd} \\ \mathbb{Z}_2 & \text{for } k \text{ odd, } 0 < k < n \\ 0 & \text{otherwise} \end{cases}$$

Example 2.43: Lens Spaces. This example is somewhat more complicated. Given an integer $m > 1$ and integers $\ell_1, \dots, \ell_n$ relatively prime to m , define the **lens space** $L = L_m(\ell_1, \dots, \ell_n)$ to be the orbit space $S^{2n-1}/\mathbb{Z}_m$ of the unit sphere $S^{2n-1} \subset \mathbb{C}^n$ with the action of $\mathbb{Z}_m$ generated by the rotation $\rho(z_1, \dots, z_n) = (e^{2\pi i \ell_1/m} z_1, \dots, e^{2\pi i \ell_n/m} z_n)$, rotating the j^{th} $\mathbb{C}$ factor of $\mathbb{C}^n$ by the angle $2\pi \ell_j/m$. In particular, when $m = 2$, ρ is the antipodal map, so $L = \mathbb{RP}^{2n-1}$ in this case. In the general case, the projection $S^{2n-1} \rightarrow L$ is a covering space since the action of $\mathbb{Z}_m$ on S^{2n-1} is free: Only the identity element fixes any point of S^{2n-1} since each point of S^{2n-1} has some coordinate z_j nonzero and then $e^{2\pi i k \ell_j/m} z_j \neq z_j$ for $0 < k < m$, as a result of the assumption that ℓ_j is relatively prime to m .

We shall construct a CW structure on L with one cell e^k for each $k \leq 2n - 1$ and show that the resulting cellular chain complex is

$$0 \rightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{m} \mathbb{Z} \xrightarrow{0} \dots \xrightarrow{0} \mathbb{Z} \xrightarrow{m} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0$$

with boundary maps alternately 0 and multiplication by m . Hence

$$H_k(L_m(\ell_1, \dots, \ell_n)) = \begin{cases} \mathbb{Z} & \text{for } k = 0, 2n - 1 \\ \mathbb{Z}_m & \text{for } k \text{ odd, } 0 < k < 2n - 1 \\ 0 & \text{otherwise} \end{cases}$$